

Name: _____

Date: _____

VOLUME AND SURFACE AREA, (SURFACE AREA OF CYLINDERS AND SPHERES)

LO: I can calculate the surface area of cylinders and spheres using formulae.

Surface area of cylinders and spheres

The **surface area** of a 3D shape is the total area of all its faces. A cylinder has two circular ends and a curved surface that unfolds into a rectangle. A **sphere** has a single curved surface.

Formulae: Cylinder SA = $2\pi r^2 + 2\pi rh$. Sphere SA = $4\pi r^2$. Always include units² in your answer.

Quick examples

●	Cylinder: $2\pi r^2 + 2\pi rh$	two circles plus curved surface
●	Sphere: $4\pi r^2$	four times the area of a great circle
●	Curved surface of cylinder = $2\pi rh$	rectangle when unrolled
●	Cylinder SA = $\pi r^2 h$	that is the volume formula

Key idea

Surface area is the total area covering the outside of a 3D shape. Use the correct formula for each shape.

$$\text{Cylinder SA} = 2\pi r^2 + 2\pi rh$$

Two circular ends plus the curved surface area

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - cylinder surface area

Find the surface area of a cylinder with radius 5 cm and height 8 cm. Give your answer to 1 d.p.

$$\begin{aligned}\text{Two circles: } 2\pi r^2 &= 2 \times \pi \times 5^2 = 50\pi \\ \text{Curved surface: } 2\pi rh &= 2 \times \pi \times 5 \times 8 \\ &= 80\pi \\ \text{Total} &= 50\pi + 80\pi = 130\pi = 408.4 \text{ cm}^2\end{aligned}$$

$$408.4 \text{ cm}^2$$

Add the area of both circles and the curved surface.

Example 2 - sphere surface area

Find the surface area of a sphere with radius 6 cm. Give your answer to 1 d.p.

$$\begin{aligned}\text{SA} &= 4\pi r^2 \\ &= 4 \times \pi \times 6^2 \\ &= 144\pi = 452.4 \text{ cm}^2\end{aligned}$$

$$452.4 \text{ cm}^2$$

Substitute the radius into $4\pi r^2$.

Example 3 - hemisphere surface area

Find the total surface area of a solid hemisphere with radius 4 cm. Give your answer to 1 d.p.

$$\begin{aligned}\text{Curved surface} &= \text{half sphere} = 2\pi r^2 \\ &= 2 \times \pi \times 16 = 32\pi \\ \text{Flat circle} &= \pi r^2 = 16\pi \\ \text{Total} &= 32\pi + 16\pi = 48\pi = 150.8 \text{ cm}^2\end{aligned}$$

$$150.8 \text{ cm}^2$$

A solid hemisphere has a curved part plus a flat circular base.

Example 4 - cylinder with no top

A tin can (open top) has radius 3 cm and height 10 cm. Find the surface area to 1 d.p.

$$\begin{aligned}\text{One circle (base): } \pi r^2 &= 9\pi \\ \text{Curved surface: } 2\pi rh &= 2 \times \pi \times 3 \times 10 = 60\pi \\ \text{Total} &= 9\pi + 60\pi = 69\pi = 216.8 \text{ cm}^2\end{aligned}$$

$$216.8 \text{ cm}^2$$

Open-top means only one circular end is included.

YOUR TURN – PRACTICE (deliberate practice)

1. Cylinder surface area

Find the total surface area of each cylinder. Round to 1 d.p.

- a) $r = 3 \text{ cm}$, $h = 7 \text{ cm}$
- b) $r = 4 \text{ cm}$, $h = 10 \text{ cm}$
- c) $r = 2 \text{ cm}$, $h = 9 \text{ cm}$
- d) $r = 6 \text{ cm}$, $h = 5 \text{ cm}$

2. Sphere surface area

Find the surface area of each sphere. Round to 1 d.p.

- a) $r = 5 \text{ cm}$
- b) $r = 3 \text{ cm}$
- c) $r = 10 \text{ cm}$
- d) $r = 7 \text{ cm}$

3. Hemispheres

Find the total surface area of each solid hemisphere. Round to 1 d.p.

- a) $r = 5 \text{ cm}$
- b) $r = 8 \text{ cm}$
- c) $r = 2 \text{ cm}$
- d) $r = 6 \text{ cm}$

4. Open cylinders

Find the surface area of each open-top cylinder. Round to 1 d.p.

- a) $r = 4 \text{ cm}$, $h = 6 \text{ cm}$
- b) $r = 5 \text{ cm}$, $h = 12 \text{ cm}$
- c) $r = 3 \text{ cm}$, $h = 8 \text{ cm}$
- d) $r = 7 \text{ cm}$, $h = 4 \text{ cm}$

5. Mixed problems

Solve each surface area problem. Round to 1 d.p.

- a) Cylinder: $r = 2.5 \text{ cm}$, $h = 6 \text{ cm}$. Total SA?
- b) Sphere with diameter 14 cm . SA?
- c) Hemisphere: $r = 9 \text{ cm}$. Total SA?
- d) Cylinder with no ends: $r = 4 \text{ cm}$, $h = 10 \text{ cm}$. Curved SA?
- e) Sphere with circumference 31.4 cm . Find r , then SA.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Cylinder $SA = 2\pi r^2 + 2\pi rh$ includes both circles and the curved surface ☐

Correct answer: _____

Reason: _____

b) Sphere $SA = 2\pi r^2$ ☐

Correct answer: _____

Reason: _____

c) The curved surface area of a cylinder equals $2\pi rh$ ☐

Correct answer: _____

Reason: _____

d) A hemisphere has surface area exactly half of a full sphere ☐

Correct answer: _____

Reason: _____

e) Surface area is always measured in square units ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cylindrical water tank has radius **1.2 m** and height **2 m**. The outside (including top and base) needs painting. Paint covers **8 m² per litre** and costs **£7.50 per litre**. Find the total cost of paint needed.

Expression: _____

Simplified: _____

b) A solid shape is made from a hemisphere of radius **5 cm** joined to a cylinder of the same radius and height **8 cm**. Find the total surface area of the shape. (The join between hemisphere and cylinder is not an external surface.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

